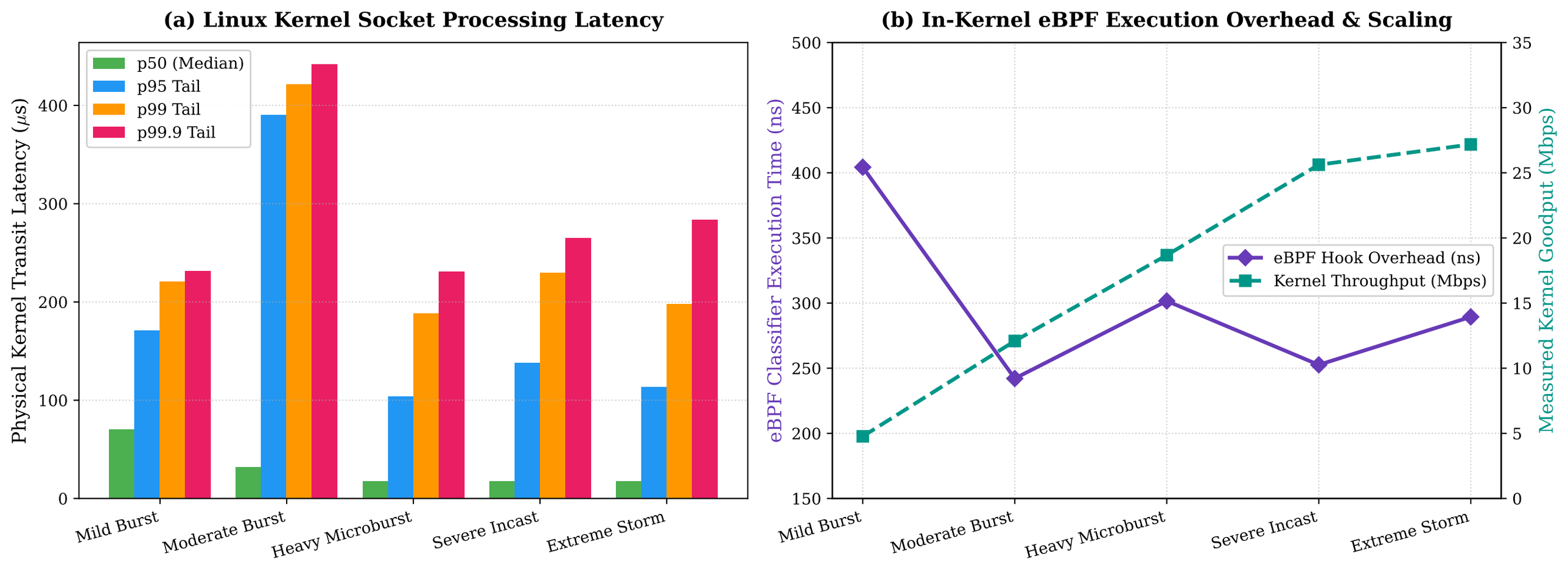
(a) Linux Kernel Socket Processing Latency
Physical Kernel Transit Latency (μs)
p50 (Median)
p95 Tail
p99 Tail
p99.9 Tail
0
100
200
300
400
Mild Burst
Moderate Burst
Heavy Microburst
Severe Incast
Extreme Storm
(b) In-Kernel eBPF Execution Overhead & Scaling
eBPF Classifier Execution Time (ns)
Measured Kernel Goodput (Mbps)
eBPF Hook Overhead (ns)
Kernel Throughput (Mbps)
150
200
250
300
350
400
450
500
0
5
10
15
20
25
30
35
Mild Burst
Moderate Burst
Heavy Microburst
Severe Incast
Extreme Storm